OAK INTERNATIONAL SCHOOL

Deliverable 5: Harmonised Scope and Sequence Plans (Classes CM2 to Première)

Document Reference: OIS-MTH-SCQ-2026-V1.0

Curricular Scope: 7 Academic Grade Levels (CM2/MYP 1 to Première/DP 2)

Standard Academic Year Model: Three Instructional Terms (Term 1: September–December; Term 2: January–March; Term 3: April–June)

Important Note: All instructional time allocations (hours/weeks) and unit sequences are institutional pedagogical recommendations designed by our curriculum team to harmonise both syllabi; they do not represent statutory state timetables.

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1. CLASS LEVEL 1: CM2 / MYP 1

Term 1: Number Sense, Operations & Geometric Foundations (12 Weeks)

- **Unit 1.1: Whole Numbers, Place Value and the Four Operations**
- *Proposed Time*: 4 weeks (20 hours).
- *IB Content & Objectives*: Place value up to millions, standard operations, mental estimation, solving word problems (Criterion A & D).
- *Benin APC Competencies*: Nombres entiers jusqu'au milliard, techniques opératoires complètes, division euclidienne avec reste, calcul mental rapide.
- *Shared Conceptual Foundation*: Base-10 positional number system and distributive property of multiplication.
- *Programme-Specific Extensions*: IB: Criterion B pattern investigations on square numbers. Benin: Word problems with large currency amounts (CFA Francs).
- *Prerequisites*: Multi-digit addition and multiplication tables.
- *Bridging Lessons*: Aligning decimal point notation (12.5) and decimal comma (12,5).
- *Bilingual Vocabulary*: Place value / *Valeur de position*; Quotient / *Quotient*; Remainder / *Reste*; Dividend / *Dividende*; Divisor / *Diviseur*.
- *Common Misconceptions*: Forgetting the placeholder zero in long multiplication; ignoring the remainder in division.
- *Formative Assessments*: Weekly 10-minute mental arithmetic sprint; error-analysis exit tickets.
- *Assessment Connections*: IB Criterion A & D; Benin C_m & C_o.
- *Proposed Workbook Activities*: "The Cotonou Wholesaler Ledger": balancing commercial supply receipts.
- **Unit 1.2: Fractions and Rational Concepts**
- *Proposed Time*: 4 weeks (20 hours).
- *IB Content & Objectives*: Fractions as division, equivalent fractions, comparing fractions, adding/subtracting like denominators.
- *Benin APC Competencies*: Écriture fractionnaire, simplification, fractions égales, somme et différence de fractions de dénominateurs multiples.
- *Shared Conceptual Foundation*: Equivalence of parts of a partitioned unit.
- *Programme-Specific Extensions*: IB: Visual fraction bar models. Benin: Fraction of a whole quantity (e.g., 3/4 of 240\text{ kg}).
- *Prerequisites*: Factors, multiples, division facts.
- *Bilingual Vocabulary*: Numerator / *Numérateur*; Denominator / *Dénominateur*; Equivalent fraction / *Fraction équivalente*; Irreducible / *Irréductible*.
- *Common Misconceptions*: Adding denominators together (\frac{a}{b} + \frac{c}{b} = \frac{a+c}{2b}).
- *Assessment Connections*: IB Criterion A; Benin C_o.
- *Proposed Workbook Activities*: "Sharing the Harvest": fractional distribution of corn crops in Parakou.
- **Unit 1.3: Planar Geometry: Lines, Angles, and Polygons**
- *Proposed Time*: 4 weeks (20 hours).
- *IB Content & Objectives*: Polygons, angle types, protractor measurement, line symmetry.

- *Benin APC Competencies*: Droites parallèles et perpendiculaires, angles (droit, aigu, obtus, plat), triangles, quadrilatères particuliers, médiatrice.
- *Shared Conceptual Foundation*: Geometric invariance under translation and reflection.
- *Programme-Specific Extensions*: IB: Criterion C geometric shape profiles. Benin: Compass-and-straightedge construction protocols.
- *Prerequisites*: Use of a ruler and pencil.
- *Bilingual Vocabulary*: Perpendicular / *Perpendiculaire*; Parallel / *Parallèle*; Perpendicular bisector / *Médiatrice*; Acute / *Aigu*; Obtuse / *Obtus*.
- *Common Misconceptions*: Confusing the inner and outer scale of a circular protractor.
- *Assessment Connections*: IB Criterion C; Benin C_m.
- *Proposed Workbook Activities*: "Architectural Plans of the Royal Palaces of Abomey": drawing verified orthogonal layouts.

Term 2: Decimals, Metric Mensuration & Proportionality (11 Weeks)

- **Unit 1.4: Decimals, Metric System and Conversions**
- *Proposed Time*: 4 weeks (20 hours).
- *Shared Foundation*: Decimal place value and metric conversion prefixes (kilo-, hecto-, deca-, deci-, centi-, milli-).
- *Extensions*: Benin: Agrarian units (ha, a, ca) and volume-capacity-mass equivalence (1\text{ dm}^3 = 1\text{ L} = 1\text{ kg}).
- *Workbook Activity*: "Water Reservoir Calibration in Dassa-Zoumè".
- **Unit 1.5: Perimeter, Area, and 3D Solids**
- *Proposed Time*: 4 weeks (20 hours).
- *Shared Foundation*: Perimeter of polygons, area formulas (square, rectangle, triangle, circle), volume of rectangular prisms (V = L \times w \times h).
- *Extensions*: IB: Packing unit cubes investigation (Criterion B). Benin: Trapezius and rhombus area formulas for CEP.
- *Workbook Activity*: "Fencing and Paving a Market Stall in Dantokpa".
- **Unit 1.6: Proportionality, Percentages and Commercial Maths**
- *Proposed Time*: 3 weeks (15 hours).
- *Shared Foundation*: Multiplicative scaling, percentages (10\%, 25\%, 50\%).
- *Extensions*: Benin: Commercial arithmetic formulas (*Prix d'achat, Frais, Prix de revient, Prix de vente, Bénéfice, Perte*).
- *Workbook Activity*: "The Songhaï Organic Farm Budget".

Term 3: Coordinate Grids, Data, Probability & Exam Prep (9 Weeks)

- **Unit 1.7: First Quadrant Coordinates and Statistics** (3 weeks / 15 h).
- **Unit 1.8: Probability Foundations & Patterns** (2 weeks / 10 h) [IB Focus].
- **Unit 1.9: Integrated CEP & MYP 1 Capstone Review** (4 weeks / 20 h).

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2. CLASS LEVEL 2: SIXIÈME / MYP 2

Term 1: Number Systems, Prime Factors & Synthetic Geometry (12 Weeks)

- **Unit 2.1: Divisibility, Prime Factorisation, HCF & LCM** (3 weeks / 15 h).
- *Vocabulary*: Prime factor / *Facteur premier*; Divisibility / *Divisibilité*; LCM / *PPCM*; HCF (GCD) / *PGCD*.
- **Unit 2.2: Directed Numbers (\mathbb{Z}) [Bridging Unit for MYP 2]** (3 weeks / 15 h).
- *Bridging Note*: Explicitly introduces negative integers to satisfy MYP 2 early requirements while maintaining Beninese focus on signed magnitude.
- **Unit 2.3: Synthetic Plane Geometry & Geometric Sets** (3 weeks / 15 h).
- *Benin Focus*: (SA 1) Notions ensemblistes (\in, \notin, \cap, \cup), segments, rays, perpendicular bisectors.
- **Unit 2.4: Axial Symmetry (Line Reflection)** (3 weeks / 15 h).
- *Shared Core*: Reflection across straight lines, conservation of distances, angles, and areas.

Term 2: Fractions, Algebraic Intro & 3D Mensuration (11 Weeks)

- **Unit 2.5: Fractions: Multiplication, Division & Operations** (4 weeks / 20 h).
- **Unit 2.6: Algebraic Expressions & One-Step Equations** (3 weeks / 15 h).

- **Unit 2.7: Cuboids, Cubes, Prisms: Nets, Area & Volume** (4 weeks / 20 h).
- *Benin Focus*: (SA 3) Cavalier perspective, lateral and total surface area, volume formulas.

Term 3: Proportionality, Data Handling & Deductive Proof (9 Weeks)

- **Unit 2.8: Proportionality, Scale & Percentage Applications** (3 weeks / 15 h).
- **Unit 2.9: Statistics: Tables, Bar Charts & Data Interpretation** (3 weeks / 15 h).
- **Unit 2.10: Formal Geometric Proof Workshops & Year-End Review** (3 weeks / 15 h).
- *Deductive Protocol*: Mastery of « On sait que... Or si... Donc... ».

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3. CLASS LEVEL 3: CINQUIÈME / MYP 3

Term 1: Directed Numbers, Powers of 10 & Parallel Angles (12 Weeks)

- **Unit 3.1: Operations in Relative Numbers (\mathbb{Z} and Relatives)** (4 weeks / 20 h).
- *Shared Core*: Number line ordering, sign rules, parentheses removal, distance between two points.
- **Unit 3.2: Powers of 10, Index Laws & Scientific Notation** (3 weeks / 15 h).
- *Shared Core*: 10^n, 10^{-n}, index addition/subtraction, scientific notation a \times 10^k, order of magnitude.
- **Unit 3.3: Angles and Parallel Lines** (5 weeks / 25 h).
- *Shared Core*: Alternate interior, alternate exterior, and corresponding angles; vertically opposite angles; triangle angle sum (180^\circ).

Term 2: Fractions, Central Symmetry & Parallelograms (11 Weeks)

- **Unit 3.4: Fractions: Unlike Denominators & Four Operations** (3 weeks / 15 h).
- **Unit 3.5: Central Symmetry (Point Reflection)** (3 weeks / 15 h).
- *Shared Core*: Point symmetry, center of symmetry, invariant properties.
- **Unit 3.6: Parallelograms & Special Quadrilaterals** (3 weeks / 15 h).
- *Shared Core*: Diagonal bisection, opposite side equality, rhombus, rectangle, square proofs.
- **Unit 3.7: Prisms, Cylinders & Solids of Revolution** (2 weeks / 10 h).

Term 3: Proportionality, Data, Probability & Sequences (9 Weeks)

- **Unit 3.8: Proportionality, Rates & Linear Journey Graphs** (3 weeks / 15 h).
- **Unit 3.9: Statistics: Frequencies, Percentages & Circular Charts** (2 weeks / 10 h).
- **Unit 3.10: Probability Trees & Linear Sequences [IB Module]** (2 weeks / 10 h).
- **Unit 3.11: Integrated Problem-Solving Review** (2 weeks / 10 h).

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4. CLASS LEVEL 4: QUATRIÈME / MYP 4

Term 1: Pythagoras' Theorem, Rational Numbers & Exponents (12 Weeks)

- **Unit 4.1: Pythagoras' Theorem and its Converse** (4 weeks / 20 h).
- *Shared Core*: a^2 + b^2 = c^2, right-triangle identification, 2D and 3D applications.
- **Unit 4.2: Rational Numbers (\mathbb{Q}) and Sign Rules** (3 weeks / 15 h).
- **Unit 4.3: Integer Exponents and Powers** (2 weeks / 10 h).
- *Shared Core*: Rules a^m \times a^n = a^{m+n}, (a^m)^n = a^{mn}, (ab)^n = a^n b^n, a^{-n} = \frac{1}{a^n}.
- **Unit 4.4: Algebraic Expansion, Single/Double Distributivity** (3 weeks / 15 h).

Term 2: Triangle Geometry, Vectors, Translations & Equations (11 Weeks)

- **Unit 4.5: Triangle Concurrency & Midpoint Theorem** (3 weeks / 15 h).
- *Benin Focus*: Centroid G (2/3 rule), orthocentre H, circumcentre, incentre.
- **Unit 4.6: Introduction to Vectors and Translations** (3 weeks / 15 h).
- *Shared Core*: Direction, sense, norm; Chasles' relation \vec{AB} + \vec{BC} = \vec{AC}; column vectors \begin{pmatrix} x \\ y \end{pmatrix}.
- **Unit 4.7: Linear Equations in One Unknown** (3 weeks / 15 h).
- *Shared Core*: Solving ax + b = cx + d, fractional equations, word problem translation.
- **Unit 4.8: Pyramids and Right Cones: Surface & Volume** (2 weeks / 10 h).

Term 3: Trigonometry, Statistics & Cumulative Data (9 Weeks)

- **Unit 4.9: Right-Angled Trigonometry (\cos, \sin, \tan)** (4 weeks / 20 h).
- *Harmonised Bridge*: Teach \sin and \tan alongside Benin's prescribed \cos.
- **Unit 4.10: Cumulative Statistics, Boxplots & Linear Functions** (3 weeks / 15 h).
- **Unit 4.11: Integrated Application & Criterion B/D Investigation** (2 weeks / 10 h).

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5. CLASS LEVEL 5: TROISIÈME / MYP 5

Term 1: Thales' Theorem, Real Numbers & Algebraic Identities (12 Weeks)

- **Unit 5.1: Thales' Intercept Theorem & Converse** (4 weeks / 20 h).
- *Shared Core*: Intercept ratios \frac{AM}{AB} = \frac{AN}{AC} = \frac{MN}{BC}, similarity, scale factor k.
- **Unit 5.2: The Real Number System (\mathbb{R}) & Radical Simplification** (4 weeks / 20 h).
- *Shared Core*: Surds, exact calculations, conjugate expressions in denominators.
- **Unit 5.3: Algebraic Identities & Complete Factorisation** (4 weeks / 20 h).
- *Shared Core*: (a\pm b)^2, a^2-b^2, null-factor equations (ax+b)(cx+d)=0, rational fractions.

Term 2: Trigonometry, Coordinate Geometry, Vectors & Linear Systems (11 Weeks)

- **Unit 5.4: Trigonometric Relations & Special Angles** (3 weeks / 15 h).
- *Shared Core*: Fundamental identities \sin^2\theta + \cos^2\theta = 1, \tan\theta = \frac{\sin\theta}{\cos\theta}, values for 30^\circ, 45^\circ, 60^\circ.
- **Unit 5.5: Coordinate Geometry & Vectors in Orthonormal Systems** (3 weeks / 15 h).
- *Shared Core*: Vector coordinates, midpoint, distance formula, collinearity (xy'-x'y=0), orthogonality (xx'+yy'=0).
- **Unit 5.6: Affine Functions & Straight Line Equations** (3 weeks / 15 h).
- *Shared Core*: y = mx + p, slope calculation, parallel and perpendicular slopes.
- **Unit 5.7: Linear Systems (2 \times 2) & Problem Formulation** (2 weeks / 10 h).

Term 3: Circle Geometry, 3D Frustums, Statistics & Dual Exam Clinics (9 Weeks)

- **Unit 5.8: Inscribed Angles, Circle Geometry & 3D Frustums** (2 weeks / 10 h).
- **Unit 5.9: Grouped Continuous Statistics & Histograms** (2 weeks / 10 h).
- **Unit 5.10: Sine/Cosine Rules & Quadratic Formula [IB Extension]** (2 weeks / 10 h).
- **Unit 5.11: Intensive BEPC & MYP 5 Dual Examination Practicum** (3 weeks / 15 h).

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6. CLASS LEVEL 6: SECONDE / DP 1

Term 1: Real Numbers, Quadratics & Functional Analysis (12 Weeks)

- **Unit 6.1: Subsets of \mathbb{R}, Intervals, and Absolute Value** (3 weeks / 15 h).
- **Unit 6.2: Quadratic Trinomials, Discriminant \Delta & Inequations** (4 weeks / 20 h).
- *Shared Core*: \Delta = b^2 - 4ac, roots, factorisation, vertex form, sign tables.
- **Unit 6.3: General Concept of Function & Graph Transformations** (5 weeks / 25 h).
- *Shared Core*: Domain D_f, parity, monotonicity, extrema, reference functions, transformations y = af(b(x-c))+d.

Term 2: 2D Vector Spaces, Dot Product & Radian Trigonometry (11 Weeks)

- **Unit 6.4: Vector Spaces, Bases & Cartesian Equations of Lines** (3 weeks / 15 h).
- **Unit 6.5: Scalar Product (Dot Product) & Metric Relations** (4 weeks / 20 h).
- *Shared Core*: Dot product definition, coordinates (xx'+yy'), Al-Kashi theorem, median theorem, circle equations.
- **Unit 6.6: Radian Measure, Unit Circle & Trigonometric Identities** (4 weeks / 20 h).
- *Shared Core*: Radians, circular functions, addition and double-angle formulas, exact values.

Term 3: Sequences, Logarithms, 3D Space & Statistics (9 Weeks)

- **Unit 6.7: Arithmetic & Geometric Sequences and Series [DP 1 Core]** (3 weeks / 15 h).
- **Unit 6.8: 3D Incidence Geometry & Orthogonality [Benin 2nde C/D Core]** (2 weeks / 10 h).
- **Unit 6.9: Introductory Logarithms & Exponential Equations [DP 1 Core]** (2 weeks / 10 h).
- **Unit 6.10: Univariate & Bivariate Statistics + GDC Practicum** (2 weeks / 10 h).

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7. CLASS LEVEL 7: PREMIÈRE / DP 2

Term 1: Differential Calculus, Curve Sketching & Sequences (12 Weeks)

- **Unit 7.1: Differential Calculus: Derivative at a Point & Tangents** (4 weeks / 20 h).
- *Shared Core*: Difference quotient, derivative function, power/product/quotient rules, chain rule.
- **Unit 7.2: Limits, Asymptotes & Complete Functional Curve Sketching** (4 weeks / 20 h).
- *Shared Core*: Indeterminate forms, vertical/horizontal/oblique asymptotes, variation tables, graphing.
- **Unit 7.3: Numerical Sequences: Monotonicity, Bounds & Convergence** (4 weeks / 20 h).
- *Shared Core*: Arithmetic and geometric sequences, sum formulas, financial modelling.

Term 2: Barycentres, Combinatorics, Probability & Trigonometry (11 Weeks)

- **Unit 7.4: Barycentres of Weighted Points & Level Curves [Benin 1ère C/D Priority]** (4 weeks / 20 h).
- *Benin Core*: Existence condition \sum \alpha_i \neq 0, reduction \sum \alpha_i \vec{MA_i} = (\sum \alpha_i)\vec{MG}, partial barycentres, level curves.
- **Unit 7.5: Combinatorics, Permutations & Combinations** (3 weeks / 15 h).
- *Shared Core*: p-lists, arrangements A_n^p, combinations \binom{n}{p}, Pascal's triangle, Binomial theorem.
- **Unit 7.6: Probability Distributions & Discrete Random Variables** (2 weeks / 10 h).
- *Shared Core*: Conditional probability, independence, probability distributions, E(X), \text{Var}(X), Binomial distribution.
- **Unit 7.7: Advanced Trigonometric Equations & Simpson Formulas** (2 weeks / 10 h).

Term 3: Integration, DP Internal Assessment & Baccalauréat Clinics (9 Weeks)

- **Unit 7.8: Integral Calculus: Anti-derivatives & Definite Integrals [DP 2 Core]** (3 weeks / 15 h).
- **Unit 7.9: Linear Programming via Half-Planes [Benin 1ère Core]** (2 weeks / 10 h).
- **Unit 7.10: DP Internal Assessment (Exploration) Finalisation** (2 weeks / 10 h).
- **Unit 7.11: Baccalauréat Première & DP Mock Examination Practicum** (2 weeks / 10 h).

End of Deliverable 5 — Harmonised Scope and Sequence Plans.